

Tarun Reddy

SOFTWARE ENGINEER

tarunlakkireddy7@gmail.com | +1 (572) 205-7781 | Oklahoma City, OK – 73034

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

Software Engineer with 6+ years of experience designing, developing, and deploying scalable distributed applications and cloud-native services across banking, retail, supply chain, and enterprise domains. Strong expertise in Java, Python, Spring Boot, Microservices, REST APIs, React.js, Apache Kafka, AWS, Docker, and Kubernetes, with hands-on experience building production systems, event-driven architectures, and high-throughput services. Experienced in Generative AI, Large Language Models (LLMs), prompt engineering, NLP-oriented applications, vector search, AI application development, and intelligent content processing through hands-on project implementation. Strong understanding of system design, distributed processing, reliability, scalability, API architecture, CI/CD, MLOps-oriented deployment practices, and SDLC. Experienced in partnering with engineering, QA, DevOps, product, and business teams to take applications from design and development through testing, deployment, monitoring, and production operations.

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3

GenAI & Machine Learning: Generative AI, Large Language Models (LLMs), Prompt Engineering, NLP, AI Application Development, Vector Search, RAG, Machine Learning

AI & Data Technologies: LangChain, Vector Databases, Embeddings, Text Processing, Model Integration, Data Processing, Feature Extraction, AI Workflow Automation

Frameworks & Distributed Systems: Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, React.js, REST APIs, Microservices

Messaging & Streaming: Apache Kafka, RabbitMQ, Redis, Event-Driven Architecture, Asynchronous Processing, Streaming Workflows

Cloud & MLOps: AWS, EC2, S3, RDS, Lambda, API Gateway, IAM, CloudWatch, Docker, Kubernetes, Jenkins, Terraform

Databases & Data Processing: PostgreSQL, MySQL, Oracle, MongoDB, DynamoDB, SQL Optimization, Database Design, Data Modeling

Engineering & SDLC: Git, Maven, CI/CD, JUnit, Mockito, Postman, SonarQube, Agile, Scrum, SDLC, Code Reviews

WORK EXPERIENCE

Senior Software Engineer | PNC Financial, USA

Feb 2026 – Present

- Developed scalable Java, Spring Boot, and React.js applications and REST APIs supporting customer onboarding, account management, payments, transaction processing, and secure self-service workflows across enterprise banking platforms.
- Engineered distributed microservices for payment processing, account validation, transaction orchestration, reconciliation, fraud screening, and audit reporting, applying scalable system design and reliability patterns to support high-volume business operations.
- Built event-driven and streaming architectures using Apache Kafka and RabbitMQ for real-time transaction processing, payment notifications, fraud alerts, and asynchronous communication across distributed services.
- Applied secure API and application architecture using Spring Security, OAuth 2.0, JWT, RBAC, SSL/TLS, and API gateways to protect customer data and financial transactions across distributed enterprise systems.
- Tuned application performance through Redis coaching, SQL optimization, connection pooling, and database tuning across Oracle and PostgreSQL, improving API response times and transaction throughput during peak workloads.
- Automated software delivery through Git, Jenkins, Docker, Kubernetes, Maven, and SonarQube, integrating automated testing, code-quality validation, build processes, and deployment workflows across development, QA, and production environments.
- Deployed cloud-native distributed applications on AWS EC2, RDS, Lambda, S3, IAM, API Gateway, and CloudWatch, supporting scalable services, high availability, operational monitoring, and production reliability.
- Established centralized logging, monitoring, and observability using ELK Stack, Prometheus, Grafana, Splunk, and AppDynamics, enabling production troubleshooting, service-level monitoring, transaction traceability, and faster incident resolution.

Software Developer Intern | Coforge, USA

May 2025 – Nov 2025

- Assisted in developing an enterprise HRMS and Payroll Management System using Java, Spring Boot, Hibernate, and REST APIs, supporting employee onboarding, attendance, payroll processing, leave management, timesheets, and performance workflows.
- Developed React.js and JavaScript frontend modules with MySQL-backed services, implementing employee self-service capabilities and integrating REST APIs with backend business logic and enterprise application workflows.
- Partnered with senior developers in Agile/Scrum SDLC to resolve application defects, optimize SQL queries, support Git/Jenkins CI/CD workflows, perform unit testing, participate in code reviews, and maintain technical documentation for reliable feature delivery.

Software Engineer | Ojas Innovative Technologies, India

Jan 2021 – Dec 2023

- Developed enterprise financial management applications using Java, Spring Boot, Hibernate, React.js, and REST APIs, supporting billing, payment processing, account management, invoicing, reconciliation, and financial reporting across high-volume business operations.

- Architected scalable RESTful APIs and Spring Boot microservices for payment workflows, invoice generation, customer account services, transaction processing, and financial settlement, applying distributed system and service-oriented design principles.
- Implemented event-driven processing with Apache Kafka and RabbitMQ for real-time payment notifications, transaction validation, reconciliation events, and asynchronous financial workflows, improving service scalability and processing reliability.
- Delivered responsive React.js dashboards using JavaScript, TypeScript, HTML5, and CSS3, providing real-time visibility into payment status, account balances, transaction history, and operational KPIs for business users.
- Improved PostgreSQL and MySQL performance through query tuning, indexing, transaction optimization, and data-access improvements, strengthening reporting performance and high-volume transaction processing.
- Integrated Redis caching and session management to accelerate frequently accessed data retrieval, reduce database load, and improve application responsiveness during high-traffic processing periods.
- Utilized AWS EC2, S3, Lambda, IAM, SQS, SNS, and API Gateway to implement scalable cloud-native services supporting payment integrations, document processing, asynchronous workflows, and distributed application communication.
- Streamlined build, testing, and deployment processes using Git, Jenkins, Docker, and Kubernetes, establishing repeatable CI/CD workflows and consistent application delivery across development, QA, and production environments.
- Worked with business analysts, product owners, QA engineers, DevOps teams, and finance stakeholders in an Agile/Scrum environment to troubleshoot production issues, conduct code reviews, improve system reliability, and deliver enhancements to distributed financial applications.

Software Developer | Lumen Technologies | India

Mar 2019 – Dec 2020

- Developed Java-based web applications and RESTful APIs supporting a large-scale shipping and logistics platform, enabling shipment tracking, order processing, warehouse operations, and customer-facing logistics services.
- Built responsive React.js interfaces using JavaScript, TypeScript, HTML5, and CSS3, delivering operational dashboards, shipment status capabilities, and user-facing workflow enhancements.
- Implemented backend services and business logic using Java, Spring MVC, Servlets, and Hibernate, supporting shipment processing, inventory management, order fulfillment, and integrations with enterprise logistics systems.
- Optimized PostgreSQL and MySQL queries, stored procedures, and database operations to improve application performance, logistics reporting, shipment analytics, and backend data processing.
- Troubleshot production application issues, performed root-cause analysis, and implemented corrective fixes to maintain availability and reliable processing across the logistics platform.
- Supported AWS-based application deployment using EC2 and S3, assisting with environment configuration, application hosting, deployment activities, and production maintenance.
- Coordinated with business analysts, QA engineers, UI/UX designers, and Agile development teams using Git and Scrum practices to deliver logistics platform enhancements and backend services through scheduled releases.
- Maintained technical documentation covering application architecture, REST API interfaces, deployment procedures, business workflows, and production support processes to improve knowledge sharing and operational continuity.

EDUCATION

Masters in computer science | University of Central Oklahoma, OK

Jan 2024 – Dec 2025

PROJECTS

AI Powered Cold Email Automation Platform | Tools: Python, Lang Chain, Vector DB, Streamlit, VS Code, Jupyter

Developed an end-to-end Generative AI application that uses an LLM to analyze job-posting content, extract relevant requirements, retrieve contextual portfolio information through vector search and embeddings, and generate personalized outreach emails. Implemented an LLM orchestration workflow using LangChain, prompt engineering, retrieval-based generation, and vector database technologies, providing an interactive Streamlit interface for AI-assisted content generation.

Secure Mental Health Support Platform with AI Moderation | Tools: Node.js, React Native, Typescript, Express.js, Mongo DB, VS
Developed a cross-platform application incorporating LLM-based content moderation and NLP-oriented processing for peer-to-peer support forums. Designed application workflows for medication tracking, appointment scheduling, and moderated user interactions, with emphasis on secure data handling, API integration, and reliable AI-assisted content classification and moderation.

Serverless Attendance Application | Tools: JavaScript, bootstrap, Firebase, NoSQL, cloud functions, google Job scheduler

Developed a serverless, event-driven application using AWS Lambda, Rekognition, Textract, S3, API Gateway, and DynamoDB to automate attendance processing through image-based recognition and document/text extraction. Designed asynchronous processing workflows and cloud-native services for handling image and unstructured data, with infrastructure-as-code and CI/CD automation using AWS CDK, CloudFormation, GitHub Actions, and Jenkins.

Remainder Webapp | Tools: Java, Spring Boot, Thymeleaf, MySQL, Spring Boot Scheduler, IntelliJ

Designed a task and reminder application using Java and Spring Boot, implementing scheduled and event-driven notification workflows for recurring tasks and time-based email notifications. Integrated cloud functions and job scheduling to support asynchronous processing and reliable delivery of user notifications.